Against the Coasean Singularity

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Abstract

AI commoditizes execution, so value migrates to the inputs execution cannot replace: local context, trust, and operational control. A growing literature argues that AI agents will reduce transaction costs enough to shift activity from firms to markets—a "Coasean singularity." This account treats coordination as an information-processing problem that AI makes cheap. We argue that coordination also requires *context revelation*, which leaks the scarce resource that generates value in the first place. Who owns the assets where context is born determines who else benefits from it. We formalize this in a graph economy where three institutional forms—fragmented market, platform router, and vertically integrated rollup—coordinate over the same nodes and edges. We model the hypothesis that platform intermediation amplifies leakage: the router ingests clients' operational data and uses it to improve service for competitors, producing higher total leakage than the open market despite lower nominal visibility. In parallel-world comparison, the rollup generates a mean 24% higher net surplus than the router across 10 seeds (10/10 pass). Under passive-router assumptions in a competing-world simulation, competitive feedback compounds this advantage into router extinction across all 10 tested seeds. Ownership of the scarce context, not intermediation of it, captures the value that AI creates.

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1 Introduction

Competitive advantage in a market economy comes from the abundant provision of scarce goods.[1] When a production input becomes commoditized, value migrates to whatever remains scarce. AI is commoditizing execution—the cognitive work of search, negotiation, monitoring, and coordination that previously required expensive human judgment. As execution becomes cheap, scarcity flees to the inputs that execution cannot replace: *local context* (private information about demand, capacity, and operational state that is only available at the point of origin), *trust* (relationships that generate information which would not otherwise exist), and *operational control* (the ability to act on what you know). These are the new scarce factors of production.

The standard account of AI's impact on firm structure misses this migration. It runs as follows: artificial intelligence reduces transaction costs, making it cheaper to coordinate across firm boundaries, which should—following Coase (1937)—shrink the optimal firm and expand market-based coordination. In the limit, firms dissolve into networks of autonomous AI agents transacting freely: the "Coasean singularity" (Shahidi et al., 2025). This view has broad support: Google DeepMind proposes "sandbox economies" where autonomous agents transact at speeds beyond human oversight (Tomašev et al., 2025), popular AI commentary envisions "billions of agents haggling in digital bazaars" (Shapiro, 2024), technology analysts project 60 billion agents coordinating through market protocols within a decade (Azhar, 2024), and MIT researchers propose "protocol-centric intelligence" as a replacement for firm-based coordination (Chopra et al., 2024). The thesis treats coordination as an information-processing problem and concludes that cheaper processing means less need for firms.

But if value has migrated to context, trust, and control, then the question is not how cheaply agents can process information across boundaries. It is who *owns* the scarce inputs—and what happens to those inputs when they cross a boundary. This paper argues that context, specifically, is a resource that degrades when it is shared: revealing private information to coordinate with a counterparty creates *leakage* that erodes the scarcity that made the

[1] This formulation draws on Danco (2017), who argues that capitalism at its core is "create shareholder value by providing customers with access to something scarce," and that when technology removes friction, scarcity reappears elsewhere. See also Christensen (1997) on the conservation of attractive profits.

information valuable. The severity of this leakage depends on institutional form. Ownership internalizes the context; intermediation and markets expose it.

The argument turns on four observations about context:

1. **Context revelation creates leakage.** To coordinate with a counterparty, a firm must reveal private context: demand signals, pricing sensitivity, capacity constraints, proprietary alpha. Once revealed, this information leaks to the broader market, eroding future rents. The severity of leakage depends on the institutional form through which coordination occurs.

2. **Learning compounds under common ownership.** When multiple nodes share an owner, successful coordination at one node generates learning signals that transfer to others. Cross-node learning creates a policy quality flywheel: the more nodes owned, the faster the shared policy improves, which increases the value of further acquisition.

3. **Ownership forecloses competitors' access.** In competitive dynamics, acquiring a node removes it from the router's client base, reducing the router's coordination opportunities and learning breadth. This foreclosure-through-node-removal is the mechanism that drives the competing-world tipping dynamic. Information degradation (external coordinators seeing blurred profiles of absorbed nodes) is theoretically motivated but empirically small in the parallel-world Shapley decomposition.

4. **Trust is the router's advantage, not the rollup's.** The router must *earn* trust through investment to access context that clients would not otherwise share—trust is the router's primary mechanism for getting beyond market-level information. The rollup accesses context through *ownership*, not trust: it has structural access to the full operational state by right of acquisition. When trust is removed from the model, the router loses its main differentiator while the rollup barely notices. Actuator access (execution authority) is structurally ownership-favoring ($\mu = 1.0$ for owned pairs vs. $\mu = 0.5$ for everyone else), but quantitatively minor under the current calibration (+0.005, ∼1% of the Shapley total).

We formalize these observations in a directed-graph model of economic coordination

where nodes are firms, edges are potential coordination relationships, and three institutional forms are evaluated over the same graph with different parameter regimes governing visibility, learning, trust, and actuator access.

The three institutional forms correspond to competing theses about AI-era industrial organization:

- The **market** corresponds to the Coasean singularity thesis: AI agents reduce transaction costs to near zero, making market coordination efficient enough that firms are unnecessary. This is the default prediction of transaction cost economics applied naïvely to AI.
- The **router** corresponds to the SaaS platform thesis: a platform intermediary (Stripe, Salesforce, a B2B marketplace) coordinates between independent firms using AI, capturing value through superior matching without owning the underlying assets. The router can invest in trust relationships with participants, but does not own them.
- The **rollup** corresponds to the cybernetic rollup thesis: a holding company acquires operating nodes and deploys AI to coordinate internally, capturing the surplus from reduced leakage and cross-node learning transfer. This is the thesis developed in Larson (2025) and related work by Hobart (2024).

We characterize the revelation-leakage tradeoff analytically and identify sufficient conditions for rollup dominance under the model's parameter regime. We then investigate the model computationally through simulation with Shapley value decomposition of channel contributions, multi-seed robustness, and a competing-world extension. Our main findings are:

1. In parallel-world comparison across 10 graph realizations, the rollup generates a mean **24% higher net surplus** than the router (10 of 10 seeds), driven primarily by the router's learning leakage—the systematic extraction of client operational intelligence that benefits competitors.
2. **Router learning leakage is the dominant channel** (55% of Shapley-decomposed total effect). If the hypothesis that platform intermediation amplifies leakage holds, it

is the single largest driver of ownership advantage—larger than cross-node learning and leakage protection combined.

3. In a **competing-world simulation** with a passive router (no strategic response to client loss), the parallel-world advantage compounds through competitive feedback: each acquisition strengthens the rollup while shrinking the router's client base, producing a tipping dynamic that drives the **router to extinction across all 10 tested seeds**. The rollup achieves 3–4× the router's cumulative surplus.

4. **Superlinearity emerges from structure alone**: portfolio value exceeds the sum of parts (ratio 1.48) without assumed complementarity.

5. The router is **worse than the market for leakage**: despite lower nominal visibility, the router's total leakage cost (0.72 per event) is 3× the market's (0.23), because the router extracts and redistributes operational intelligence to competitors.

The paper contributes to three literatures. First, to the theory of the firm (Coase, 1937; Williamson, 1975; Grossman and Hart, 1986; Hart and Moore, 1990), by identifying a mechanism—cross-node learning transfer under common ownership—that is distinct from classical hold-up, asset specificity, or residual control rights. Second, to the economics of platforms (Rochet and Tirole, 2003; Parker et al., 2016), by formalizing conditions under which ownership dominates intermediation. Third, to the emerging literature on AI and industrial organization (Agrawal et al., 2018; Brynjolfsson and Mitchell, 2017), by providing a model that makes testable predictions about which industries should see consolidation versus fragmentation as AI capabilities improve.

2 Related Literature

2.1 Transaction Cost Economics and the Theory of the Firm

The foundational insight of Coase (1937) is that firms exist to economize on transaction costs: when the cost of using the price mechanism exceeds the cost of internal organization, activities are brought within the firm. Williamson (1975, 1985) extended this framework by

identifying asset specificity, uncertainty, and frequency as the key determinants of governance structure, with the central prediction that asset-specific transactions migrate from markets to hierarchies.

The property rights theory of Grossman and Hart (1986) and Hart and Moore (1990) reframed the question: ownership matters because it confers residual control rights over assets, which determines bargaining power in states of the world not covered by incomplete contracts. Hart (1995) synthesized these into a theory of firm boundaries based on the allocation of control rights.

Our model departs from these frameworks in a specific way. In classical transaction cost economics, ownership reduces costs (haggling, hold-up, maladaptation). In our model, ownership changes the *information topology*: it prevents leakage of existing context (through internalization), accelerates learning (through richer signal quality and cross-node transfer), and forecloses context from competitors (through information withdrawal). Trust and actuator access also differ by institutional form, though our simulation shows these channels primarily benefit the platform intermediary rather than the owner under the current calibration. The relevant margin is not cost reduction but *value creation* through ownership-contingent information flows.

2.2 Platform Economics

Rochet and Tirole (2003) and Armstrong (2006) developed the theory of two-sided platforms, where an intermediary creates value by internalizing cross-side network externalities. Parker et al. (2016) extended this to platform ecosystems. The platform literature generally assumes that intermediation is sufficient—the platform need not own the assets it coordinates.

Our router regime operationalizes this assumption. The key finding is that platforms face a structural ceiling that is *worse* than mere redirection of information flow. When a firm uses a SaaS platform to coordinate—e.g., an AI customer service tool like Intercom's Fin—the platform ingests the firm's operational context (which conversations resolve, which escalate, what exception patterns emerge) and uses this data to improve its service for *all* clients, including the firm's competitors. This is not passive information exposure; it is systematic extraction and redistribution of operational intelligence. In our model, this manifests as

router learning leakage: the better the router learns from your coordination data, the more your competitors benefit. The router's total leakage cost therefore exceeds the market's, despite its lower nominal visibility, because the market merely exposes your signals while the router actively deploys them against you.

2.3 Conglomerates and the Diversification Discount

A large empirical literature documents a "conglomerate discount": diversified firms trade at a discount to the sum of their parts (Berger and Ofek, 1995; Lang and Stulz, 1994). The standard explanations—agency costs, inefficient internal capital markets, managerial empire-building (Jensen, 1986)—all predict that combining unrelated businesses destroys value.

Our model identifies conditions under which the opposite holds: a *conglomerate premium* emerges when owned nodes generate cross-vertical learning transfer, reduced information leakage, and trust-generated context. The critical moderator is the presence of an AI-enabled coordination layer that can actually capture these structural advantages. Without cross-node learning (our ablation shows), the premium collapses.

This suggests a resolution to the conglomerate puzzle: diversification destroys value when the parent company lacks the information-processing capacity to exploit the structural advantages of common ownership. AI provides that capacity, potentially reversing the sign of the diversification effect in industries where context is local and leakage-sensitive.

2.4 AI and Industrial Organization

Agrawal et al. (2018) frame AI as a reduction in the cost of prediction, implying that AI lowers the cost of contracting and should shift activities toward market governance. Brynjolfsson and Mitchell (2017) document productivity effects of AI adoption. Benzell and Brynjolfsson (2024) model AI's impact on firm boundaries through reduced communication costs.

The most comprehensive articulation of the market-expansion thesis is Shahidi et al. (2025), who analyze AI agents as autonomous market participants that search, negotiate, and transact on behalf of human principals. Their framework treats the firm boundary

question as resolved by Coase: by "dramatically reducing transaction costs," agents shift activity from firms to markets. They envision agents operating across platforms ("bring-your-own" agents), eliciting preferences at scale, and enabling previously impractical market designs such as comprehensive matching algorithms.

Our model identifies three gaps in this framework. First, Shahidi et al. (2025) treat information as freely processable: agents elicit preferences, compare options, and negotiate without cost to revealing context. Our model shows that context revelation creates *leakage*—disclosed information erodes future rents—and that the severity of leakage depends on institutional form. This means the transaction cost reduction that agents provide comes with a hidden cost that their framework does not account for.

Second, Shahidi et al. (2025) assume that valuable information preexists and merely needs to be elicited or discovered. Our model identifies a category of information—trust-generated context—that *only exists in trusted interactions*. Exception patterns, quality risks, and hidden operational states are surfaced through repeated, high-trust coordination that market interactions and platform intermediation cannot replicate at the same fidelity. This is not a search cost problem; it is a context creation problem.

Third, and most fundamentally, Shahidi et al. (2025) focus on who owns the *agent* (bring-your-own vs. platform-provided) but not who owns the *asset where context is born*. Our model shows that the critical ownership question is not "whose agent intermediates the transaction?" but "who owns the node that generates the context in the first place?" The answer to this question determines visibility, trust, actuator access, learning signal quality, and—at high ownership density—whether competitors have access to context at all (foreclosure). Agent ownership is a second-order question; asset ownership is first-order.

The incentive to foreclose rather than share context has a mechanism-design foundation. Bonatti et al. (2022) show that information monopolists facing competing buyers maximize revenue through *exclusive* provision of information, not democratization. Selling information to any buyer imposes negative externalities on all other buyers; the optimal mechanism exploits this by providing information exclusively rather than broadly. In our model, the rollup is precisely such an information monopolist: by owning the nodes where context is born, it controls the allocation of payoff-relevant information to downstream coordinators. The fore-

closure mechanism in our simulation—where absorbed nodes' context becomes private—is the structural analogue of Bonatti et al.'s exclusivity result. The Coasean singularity thesis implicitly assumes that context should flow freely through agent-mediated markets; mechanism design theory shows this is not incentive-compatible for the information holder.

3 Model

3.1 Environment

Consider a directed graph economy $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = |\mathcal{V}|$ nodes (firms or operating units) and edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ representing potential coordination relationships. Time is discrete: $t \in \{1, \ldots, T\}$. Nodes are partitioned into K verticals (industries or domains).

Definition 1 (Node)**.** Each node $i \in \mathcal{V}$ is characterized by:

- A vertical assignment $v_i \in \{1, \ldots, K\}$ — industry or domain
- Context vectors $\mathbf{d}_i, \mathbf{c}_i \in [0, 1]^7$ — demand profile and capacity profile, used to compute coordination fit between nodes
- Hidden alpha $\alpha_i \in [0, 1]$ — private information value; higher alpha means the node has more competitively valuable context to protect (or leak)
- Actuator strength $a_i \in [0, 1]$ — the node's capacity to execute coordination actions (e.g., fulfill orders, adjust operations)
- Exception rate $\xi_i \in [0, 1]$ — frequency of non-standard cases that complicate execution
- Quality risk $\omega_i \in [0, 1]$ — probability of operational failures during coordination

Definition 2 (Edge)**.** Each edge $(i, j) \in \mathcal{E}$ is characterized by:

- Base surplus $s_{ij} > 0$ — potential gross value from successful coordination between i and j
- Leakage sensitivity $\ell_{ij} \in [0, 1]$ — how much revealed context on this edge is exploitable by competitors (e.g., a pricing negotiation leaks more than a logistics handoff)

- Trust requirement $\tau_{ij} \in [0, 1]$ — how much trust is needed to surface valuable context on this edge. A high trust requirement means the coordination involves sensitive operational information (e.g., sharing proprietary quality data with a supplier) that parties will only reveal in trusted relationships; a low trust requirement means the coordination can proceed with publicly available information (e.g., a standard commodity purchase)

- Complementarity $\kappa_{ij} \in [0, 1]$ — exogenous synergy between the two nodes. The default coefficient is zero: we deliberately exclude assumed synergies to show that superlinearity emerges from structural channels alone (learning, leakage protection), not from baked-in complementarity. Turning on its coefficient in the match quality equation barely changes results, confirming this

- Latency cost $\lambda_{ij} > 0$ — time or friction cost of coordination

- Execution risk $\epsilon_{ij} \in [0, 1]$ — baseline probability of poor execution on this edge

3.2 Institutional Forms

Three institutional forms are evaluated over the same graph. Each is characterized by a parameter regime $\Theta = (\nu, \mu, \iota)$ governing visibility, actuator access, and interaction mode.

Remark: Parallel-World Comparison

We adopt a *parallel-world* design: the same underlying economy is run separately under each institutional form. Each regime faces identical initial conditions, opportunity draws, and context drift. This is the standard approach for mechanism comparison—it cleanly attributes performance differences to institutional form rather than to path-dependent competitive dynamics. An alternative *in-world competition* design, where the three forms coexist in a single economy and compete for nodes and surplus, would test strategic dynamics but risks conflating "better coordination model" with "won the race first." We present both: the parallel-world analysis for clean attribution (Sections 5.3–5.8) and a competing-world extension for competitive dynamics (Section 5.9).

Definition 3 (Institutional Form)**.** An institutional form is a triple $(\nu_{ij}, \mu_{ij}, \iota_{ij})$ where:

- $\nu_{ij} \in [0, 1]$ is **market visibility**: the fraction of revealed context that leaks to the broader market
- $\mu_{ij} \in [0, 1]$ is **actuator multiplier**: the degree of operational control over coordination execution
- $\iota_{ij} \in [0, 1]$ is **interaction mode**: how conducive the interaction channel is to generating trust. Owned subsidiaries sharing daily standups, integrated systems, and aligned incentives have high interaction mode ($\iota = 1.0$); a platform's standardized interface is moderate ($\iota = 0.6$); arm's-length market transactions between strangers are low ($\iota = 0.2$)

The three forms are parameterized as follows. Let $\mathcal{O}^t \subseteq \mathcal{V}$ denote the set of nodes owned by the rollup at time t.

Table 1: Institutional Form Parameters

Parameter	Market	Router	Rollup
Visibility ν_{ij}	1.0	0.85	$\begin{cases} 0.1 & i,j \in \mathcal{O}^t \\ 0.5 & \text{otherwise} \end{cases}$
Actuator μ_{ij}	0.5	0.5	$\begin{cases} 1.0 & i,j \in \mathcal{O}^t \\ 0.5 & \text{otherwise} \end{cases}$
Interaction ι_{ij}	0.2	0.6	$\begin{cases} 1.0 & i,j \in \mathcal{O}^t \\ 0.2 & \text{otherwise} \end{cases}$
Base learning rate η_{base}	0.03	0.03	0.03
Signal quality σ_q[a]	0.3	$0.5 + \text{breadth}$[b]	weighted avg[c]
Transfer bonus[d]	0	0	$\tau \cdot \lvert\{v_k : k \in \mathcal{O}^t\}\rvert / K$
Ownership cost	0	0	$c_{\text{overhead}} \cdot \lvert\mathcal{O}^t\rvert + \sum f_j^t$

[a]Signal quality determines how much of a coordination outcome's information content each regime can observe, modulating the effective learning rate.
[b]The router's signal quality includes a breadth bonus from observing all client interactions: $0.5 + \ln(1 + |\mathcal{E}_R^t|)/10$, reflecting how platforms like Stripe learn from aggregate transaction patterns.
[c]The rollup's signal quality is a weighted average of internal ($\sigma_{\text{own}} = 0.9$, full outcome observability) and external ($\sigma_{\text{mkt}} = 0.3$, price/quantity only), weighted by the fraction of events in each category.
[d]Transfer bonus: cross-vertical learning available only to the rollup. Proportional to the number of distinct verticals represented in the owned portfolio ($|\{v_k\}|$) divided by total verticals (K). A rollup owning logistics and healthcare nodes learns faster than one concentrated in a single vertical, because operational patterns transfer across domains.

The parameter values reflect the structural properties of each institutional form. The router's interaction mode ($\iota = 0.6$) is higher than the rollup's external interactions ($\iota = 0.2$) because the router is a *relationship platform*: it provides a standardized, repeated interface to all its clients (onboarding, account management, API integration), which is more conducive to trust generation than arm's-length market transactions. The rollup achieves $\iota = 1.0$ on owned pairs through structural alignment (shared owner, integrated systems), but when a subsidiary deals with an unowned company, there is no platform mediating the relationship—it is a standard market transaction. This asymmetry is why the Shapley decomposition shows

trust favoring the router: the router builds moderate trust broadly, while the rollup builds deep trust narrowly.

Assumption 1 (Identical Graph)**.** All three institutional forms operate over the same graph $\mathcal{G}$ with the same node and edge parameters. They differ only in the parameter regime Θ.

This assumption is critical: it ensures that any performance difference across regimes is attributable to the institutional form, not to the underlying economic structure.

3.3 Coordination Game

At each time step, each node i generates opportunities according to a Poisson process with rate λ_O. For each opportunity, node i selects a counterparty j from its neighbors and chooses a revelation level $r \in [0, 1]$ representing the fraction of private context disclosed.

3.3.1 Match Quality

Revelation improves coordination quality but with diminishing returns. Match quality is $m_{ij}(r) = \sigma(z_{ij})$, where $\sigma(\cdot)$ is the sigmoid function and the linear index z_{ij} aggregates the channels through which coordination quality is determined:

$$z_{ij} = \underbrace{\beta_0}_{\text{baseline}} + \underbrace{\beta_1 \ln(1 + b_r \cdot r)}_{\text{revelation}} + \underbrace{\beta_2 \cdot \hat{\tau}_{ij}}_{\text{trust context}} + \underbrace{\beta_3 \cdot q}_{\text{policy quality}} + \underbrace{\beta_4 \cdot \hat{a}_{ij}}_{\text{actuator}} + \underbrace{\beta_5 \cdot \kappa_{ij}}_{\text{complementarity}} + \underbrace{\beta_6 \cdot \phi_{ij}}_{\text{context fit}} \tag{1}$$

The logarithmic form of the revelation term ensures diminishing returns: the first units of context shared are most valuable. The remaining terms are: $\hat{\tau}_{ij}$, trust-generated context (Section 3.3.3); q, the regime's **policy quality**—accumulated institutional knowledge about coordination decisions, starting at 0.1 and improving through learning (Section 3.4.2); $\hat{a}_{ij}$, actuator access (Section 3.3.4); and ϕ_{ij}, **context fit**—how well node i's demand profile matches node j's capacity profile:

$$\phi_{ij} = \frac{\mathbf{d}_i \cdot \mathbf{c}_j}{\|\mathbf{d}_i\| \|\mathbf{c}_j\|} \tag{2}$$

The logarithmic form $\ln(1 + b_r \cdot r)$ ensures diminishing returns to revelation: the first units of context shared are most valuable.

3.3.2 Information Leakage

Revelation creates leakage governed by a scale parameter $\lambda_L = 0.8$ and a convexity exponent $\gamma = 1.5$ (leaking a lot is disproportionately more costly than leaking a little):

$$L_{ij}(r) = \underbrace{\lambda_L}_{\text{scale}} \cdot \underbrace{r^\gamma}_{\substack{\text{convex in}\\ \text{revelation}}} \cdot \underbrace{\ell_{ij}}_{\substack{\text{edge}\\ \text{sensitivity}}} \cdot \underbrace{\alpha_i}_{\substack{\text{info}\\ \text{value}}} \cdot \underbrace{\nu_{ij}}_{\substack{\text{market}\\ \text{visibility}}} \tag{3}$$

Instant leakage is the product of how much you reveal (r^γ), how exploitable this edge is (ℓ_{ij}), how valuable your information is (α_i), and how exposed your interaction is (ν_{ij}, which varies by institutional form per Table 1).

Leakage accumulates in a persistent stock S_i^t that depresses future surplus, governed by a decay rate $\rho = 0.8$ (leaked information stays exploitable for a long time):

$$S_i^{t+1} = \underbrace{\rho \cdot S_i^t}_{\substack{\text{decayed}\\ \text{prior stock}}} + \underbrace{L_{ij}^t}_{\substack{\text{new}\\ \text{leakage}}} \tag{4}$$

Total leakage cost combines the instant leakage with a stock penalty weighted by $\phi_L = 0.15$: $C_L = L_{ij} + \phi_L \cdot S_i^t$. The stock penalty means that past leakage continues to erode rents even after the interaction that caused it.

Router learning leakage. The router incurs an additional form of leakage absent from market interactions. When a firm coordinates through a platform, the platform ingests the firm's operational data—resolution patterns, exception rates, quality signals—and uses it to improve its aggregate model, which serves all clients including competitors. This cost is proportional to match quality (higher-quality coordination reveals more operationally valuable information):

$$L_{ij}^R = \underbrace{\lambda_R}_{\substack{\text{learning}\\ \text{leak rate}}} \cdot \underbrace{m_{ij}}_{\substack{\text{match}\\ \text{quality}}} \cdot \underbrace{s_{ij}}_{\substack{\text{base}\\ \text{surplus}}} \tag{5}$$

The router learning leakage rate $\lambda_R = 0.15$ scales with match quality m_{ij} (better coordination reveals more exploitable data) and base surplus s_{ij} (higher-value edges leak more valuable intelligence). This cost is added to the router's total leakage, making router interactions *more* leaky than market interactions despite lower nominal visibility—because the market merely exposes information while the router systematically extracts and redistributes it.

3.3.3 Trust-Generated Context

Trust-generated context is information that *only exists because a trusted interaction occurred.* It is not preexisting information that gets disclosed; it is new information created by the quality of the interaction channel:

Trust-generated context is $\hat{\tau}_{ij} = \sigma(z'_{ij}) \cdot \alpha_i$, where the sigmoid argument aggregates the factors that determine whether trust-sensitive information is surfaced:

$$z'_{ij} = \underbrace{\beta'_0}_{\text{threshold}} + \underbrace{\beta'_1 \cdot T^t_{ij}}_{\text{pairwise trust}} + \underbrace{\beta'_2 \cdot \iota_{ij}}_{\text{interaction mode}} + \underbrace{\beta'_3 \cdot (1 - \tau_{ij})}_{\text{edge accessibility}} + \underbrace{\beta'_4 \cdot \alpha_i}_{\text{info available}} \tag{6}$$

Pairwise trust $T^t_{ij} \in [0, 1]$ evolves over time with successful coordination—repeated positive interactions build trust. **Interaction mode** ι_{ij} captures the quality of the relationship channel (owned pairs share daily standups; platform clients interact through a standardized interface; market counterparties are strangers). **Edge accessibility** $(1 - \tau_{ij})$ means edges with low trust requirements surface context more easily. **Info available** α_i reflects how much trust-sensitive context the node has to offer. The final multiplication by α_i ensures that trust-generated context is proportional to the information actually available—a node with nothing to share generates no trust context regardless of how trusted the interaction is.

3.3.4 Actuator Access

The actuator term captures the degree to which matched coordination intent translates to execution:

The actuator term captures how much of the matched coordination intent translates to actual execution:

$$\hat{a}_{ij} = \underbrace{\mu_{ij}}_{\substack{\text{regime}\\\text{control}}} \cdot \underbrace{\frac{a_i + a_j}{2}}_{\substack{\text{node}\\\text{capability}}} \tag{7}$$

where μ_{ij} is the regime-specific actuator multiplier (1.0 for rollup-internal, 0.5 for everyone else) and a_i, a_j are the nodes' intrinsic operational capability. Higher actuator access reduces execution loss:

$$X_{ij} = \underbrace{\lambda_E}_{\text{scale}} \cdot \underbrace{\epsilon_{ij}}_{\substack{\text{edge}\\\text{risk}}} \cdot \underbrace{\frac{\xi_i + \omega_j}{2}}_{\substack{\text{operational}\\\text{fragility}}} \cdot \underbrace{(1 - \hat{a}_{ij})}_{\substack{\text{control}\\\text{gap}}} \tag{8}$$

Execution loss is the product of four factors:

- λ_E — **scale parameter** controlling the overall magnitude of execution losses
- ϵ_{ij} — **edge risk**: how inherently complex this coordination task is, independent of who performs it. A standard commodity purchase has low ϵ; integrating a custom software system with a supplier's legacy infrastructure has high ϵ
- $(\xi_i + \omega_j)/2$ — **operational fragility**: how prone these specific firms are to stumbling. Exception rate ξ_i measures how often node i encounters non-standard cases (e.g., a restaurant with dietary restriction requests on every other order); quality risk ω_j measures how often node j drops the ball on delivery (e.g., a warehouse with high damage rates). The average captures the combined fragility of the pair
- $(1 - \hat{a}_{ij})$ — **control gap**: the residual risk after actuator access is applied. An owner who can mandate process changes ($\hat{a} \approx 1$) has a near-zero control gap; a market participant who can only suggest ($\hat{a} \approx 0.25$) suffers most of the underlying risk

3.3.5 Net Surplus

The net surplus from a coordination event is:

$$\Pi_{ij}(r) = \underbrace{s_{ij} \cdot m_{ij}(r)}_{\text{gross value}} - \underbrace{\lambda_{ij}}_{\text{latency}} - \underbrace{C_L(r)}_{\text{leakage cost}} - \underbrace{X_{ij}}_{\text{execution loss}} \tag{9}$$

Gross value is the base surplus of the edge scaled by match quality—how much value this coordination could generate if fully realized. **Latency cost** λ_{ij} is the time and friction cost of coordinating across this edge. **Leakage cost** $C_L(r)$ is the cost of information exposure from revelation (Section 3.3.2). **Execution loss** X_{ij} is the value lost to operational failures. Net surplus must be positive for coordination to proceed.

3.3.6 Optimal Revelation

Each node chooses the revelation level that maximizes net surplus:

$$r^*_{ij} = \arg\max_{r \in [0,1]} \Pi_{ij}(r) \tag{10}$$

Proposition 1 (Revelation-Leakage Tradeoff)**.** *For any edge (i, j) with base surplus $s_{ij} > 0$, the optimal revelation r^*_{ij} is:*

(i) Decreasing in visibility ν_{ij} (fraction of context that leaks to market): $\partial r^*/\partial \nu < 0$

(ii) Decreasing in leakage sensitivity ℓ_{ij} (how exploitable revealed context is): $\partial r^*/\partial \ell < 0$

(iii) Increasing in base surplus s_{ij} (value of the coordination opportunity): $\partial r^*/\partial s > 0$

In words: nodes reveal less when leakage is costly (high visibility or sensitivity) and reveal more when the coordination prize is large. This is the core tradeoff of the model.

Proof. The first-order condition for interior r^* equates the marginal benefit of revelation (left side) with the marginal cost of leakage (right side):

$$\underbrace{s_{ij} \cdot \sigma'(z) \cdot \frac{\beta_1 b_r}{1 + b_r r}}_{\text{marginal benefit: better match quality}} = \underbrace{\lambda_L \gamma r^{\gamma-1} \ell_{ij} \cdot \alpha_i \cdot \nu_{ij}}_{\text{marginal cost: more leakage}}$$

The marginal benefit diminishes in r (logarithmic revelation term). The marginal cost increases in r (convex leakage, $\gamma > 1$). By the implicit function theorem, $\partial r^*/\partial \nu < 0$ because increasing visibility ν raises the marginal cost at every r; $\partial r^*/\partial \ell < 0$ by the same argument; $\partial r^*/\partial s > 0$ because increasing the coordination prize s raises the marginal benefit. □

Corollary 1 (Institutional Form and Revelation)**.** *Under the parameter regime in Table 1, the rollup's internal interactions have higher optimal revelation than the router or market, because lower visibility ($\nu = 0.1$ vs. 0.85 or 1.0) shifts the tradeoff toward more revelation. This directly implies higher match quality for internal interactions.*

Remark: Revelation at the boundary

Under the baseline calibration ($\lambda_L = 0.8$, $\gamma = 1.5$), match quality gains from revelation dominate leakage costs at every visibility level, so the optimizer selects $r = 1.0$ (full revelation) for all regimes in ∼99% of events. The revelation-leakage tradeoff is therefore *theoretical* under the baseline: it describes the correct comparative statics but does not produce regime-differentiated revelation behavior. At higher leakage intensity, the tradeoff binds and produces the predicted regime differentiation: at $\lambda_L = 5.0$, the rollup reveals fully in 85% of events, the router in 63%, and the market in 67%—consistent with Proposition 1. We retain the baseline calibration because the ownership advantage does not depend on differentiated revelation; it is driven by router learning leakage, cross-node learning, and leakage protection, all of which operate at $r = 1.0$. A calibration where the tradeoff binds would add regime-differentiated revelation as an additional channel favoring the rollup. Under the baseline, the realized regime separation comes from router learning leakage, cross-node learning, and leakage protection—not from differentiated revelation behavior.

3.4 Dynamics

3.4.1 Trust Dynamics

Pairwise trust T_{ij}^t (how much nodes i and j trust each other at time t) evolves with interaction outcomes. Trust builds at rate $\eta^+ = 0.08$ on success and decays at rate $\eta^- = 0.04$ on failure, with an ownership bonus b (= 1.2 for owned pairs, 1.0 otherwise). After each coordination event:

$$T_{ij}^{t+1} = \text{clip}\left[\underbrace{T_{ij}^{t}}_{\substack{\text{current}\\\text{trust}}} + \underbrace{\eta^{+}(1 - T_{ij}^{t}) \cdot b}_{\substack{\text{trust gain on}\\\text{success}}} \cdot \underbrace{\mathbf{1}[\Pi > 0]}_{\substack{\text{positive}\\\text{surplus?}}} - \underbrace{\eta^{-} T_{ij}^{t}}_{\substack{\text{trust loss on}\\\text{failure}}} \cdot \underbrace{\mathbf{1}[\Pi \leq 0]}_{\substack{\text{negative}\\\text{surplus?}}},\ 0,\ 1\right] \tag{11}$$

Trust grows when coordination succeeds ($\Pi > 0$) and decays when it fails ($\Pi \leq 0$), clipped to $[0, 1]$. The key parameters:

- η^{+} — **trust gain rate**: how fast trust builds after a successful interaction
- $(1 - T_{ij}^{t})$ — **diminishing returns**: trust builds slower as it approaches 1.0
- b — **ownership bonus**: $b = 1.2$ for owned pairs (structural alignment accelerates trust), $b = 1.0$ otherwise
- η^{-} — **trust decay rate**: how fast trust erodes after a failed interaction

3.4.2 Learning Dynamics

Each regime accumulates institutional knowledge about coordination: which counterparties work best, what revelation levels produce good outcomes, when to coordinate vs. wait. This knowledge is represented as a scalar **policy quality** $q^{t} \in [0, 1]$, starting at 0.1 (naive) and improving toward 1.0 (expert). Higher policy quality directly improves match quality in every coordination event (via the $\beta_3 \cdot q$ term in Eq. 1).

A critical design choice: all three regimes use the *same base learning rate* $\eta_{\text{base}} = 0.03$. Learning differences emerge endogenously from **signal quality** σ_q—how much of each coordination outcome's information content the regime can actually observe. The analogy: everyone improves at the same rate per unit of information, but ownership gives you access to richer information.

- **Market** ($\sigma_q = 0.3$): sees only whether a deal closed and at what price. Like learning to cook by reading restaurant reviews.
- **Router** ($\sigma_q \approx 0.5+$): sees richer outcome data from its platform, plus patterns across all clients. Like learning to cook by reading the kitchen's order tickets.

- **Rollup internal** ($\sigma_q = 0.9$): sees full operational outcomes—what worked, what broke, why. Like learning to cook by standing in the kitchen.

The policy quality update:

$$q^{t+1} = \underbrace{q^t}_{\substack{\text{current}\\\text{knowledge}}} + \underbrace{\eta_{\text{base}} \cdot \sigma_q}_{\substack{\text{effective}\\\text{learning rate}}} \cdot \underbrace{\bar{m}^t}_{\substack{\text{avg match}\\\text{quality}}} \cdot \underbrace{(1 - q^t)}_{\substack{\text{diminishing}\\\text{returns}}} + \underbrace{\tau_{\text{transfer}}}_{\substack{\text{cross-node}\\\text{transfer}}} \tag{12}$$

The effective learning rate is $\eta_{\text{base}} \cdot \sigma_q$: same base rate, modulated by how much you can see. The $(1 - q^t)$ term ensures diminishing returns—it's easy to improve from 0.1 to 0.5, hard to improve from 0.9 to 1.0. The signal quality σ_q is derived from each regime's information access:

$$\sigma_q = \begin{cases} \sigma_{\text{market}} = 0.3 & \text{market (price/quantity outcomes only)} \\ \sigma_{\text{router}} + \frac{b_R \cdot \ln(1+n_R^t)}{10} & \text{router (platform data + aggregate breadth)} \\ \frac{n_{\text{own}} \cdot \sigma_{\text{own}} + n_{\text{ext}} \cdot \sigma_{\text{market}}}{n_{\text{own}} + n_{\text{ext}}} & \text{rollup (weighted by interaction type)} \end{cases} \tag{13}$$

where $\sigma_{\text{own}} = 0.9$ (full internal outcome observability for owned pairs), $\sigma_{\text{router}} = 0.5$ (platform sees richer outcomes), $b_R = 1.0$ (router aggregate breadth), and n_R^t is the number of coordination events the router facilitated in period t (more client transactions means a richer aggregate learning signal). The router's structural advantage is *breadth*: it learns from all client interactions simultaneously, analogous to how Stripe learns from all merchant transactions. The rollup's advantage is *depth*: it observes full internal outcomes for owned pairs.

The transfer bonus τ_{transfer} is available only to the rollup:

$$\tau_{\text{transfer}} = \tau \cdot \frac{|\{v_k : k \in \mathcal{O}^t\}|}{K} \tag{14}$$

proportional to vertical diversity. A platform like Stripe does observe transactions across verticals and can transfer surface patterns (fraud signatures, churn signals) — this cross-vertical learning is already partially captured by the router's breadth bonus. The transfer bonus represents a deeper form of cross-vertical learning: operational patterns (exception

handling, capacity planning, process design) that require full internal access to identify and apply. A router knows *that* a client churned; an owner knows *why* and can apply the fix to a subsidiary in a different vertical with the same operational friction. The model draws the line at this depth distinction: breadth-based cross-vertical learning accrues to the router, depth-based operational transfer accrues to the rollup.

The router has an explicit trust-building investment mechanism. Each period, the router spends $I = 0.01$ per edge to build trust, but with diminishing returns governed by $k = 3.0$:

$$\Delta T_{ij}^{\text{router}} = \frac{\overbrace{I}^{\text{investment}}}{\underbrace{1 + C_{ij}^{\text{cumul}}/k}_{\text{diminishing returns}}} \tag{15}$$

As cumulative past investment C_{ij}^{cumul} grows, each additional dollar of trust investment yields less trust gain—the early relationship-building is cheap, but sustaining deep trust gets progressively more expensive. This ensures the router is not a strawman (it *can* build trust) while reflecting the real cost of earning trust without structural ownership.

3.4.3 Ownership Costs

Ownership is not free. The rollup incurs two categories of cost:

$$C_{\text{own}}^{t} = \underbrace{|\mathcal{O}^t| \cdot c_{\text{overhead}}}_{\text{recurring management cost}} + \underbrace{\sum_{j \in \mathcal{O}^t} f_j^t}_{\text{integration friction}} \tag{16}$$

where c_{overhead} is per-node per-period management overhead (governance, reporting, coordination bureaucracy) and f_j^t is integration friction for node j, which is set to f_0 upon acquisition and decays geometrically: $f_j^{t+1} = \delta \cdot f_j^t$. This captures the empirical regularity that acquisitions impose temporary productivity drag that diminishes as integration completes.

The rollup's *net* surplus is therefore gross coordination surplus minus ownership costs:

$$\Pi_{\text{rollup}}^{\text{net}} = \sum_{(i,j)} \Pi_{ij} - \sum_{t=1}^{T} C_{\text{own}}^{t} \tag{17}$$

This ensures the rollup must generate enough coordination surplus to justify the overhead

of ownership—a constraint absent from models that assume ownership is costless.

3.4.4 Context Foreclosure

When the rollup absorbs a node, it withdraws that node's operational context from the external information pool. This is not a portfolio-level penalty but a *microstructure-level* degradation: external coordinators see noisy versions of the absorbed node's demand and capacity profiles, because the rollup has internalized the information that was previously observable.

Formally, when a non-rollup regime tries to assess how well node j's capacity matches node i's demand, it does not see node j's true capacity profile $\mathbf{c}_j$. It sees a noisy version, with noise governed by the foreclosure scale parameter $\sigma_{\text{fc}} = 0.4$:

$$\underbrace{\tilde{\mathbf{c}}_j}_{\substack{\text{partially}\\ \text{hidden}}} = (1 - \underbrace{f}_{\substack{\text{blend}\\ \text{frac}}}) \cdot \underbrace{\mathbf{c}_j}_{\substack{\text{true}\\ \text{capacity}}} + \underbrace{f}_{\substack{\text{blend}\\ \text{frac}}} \cdot \underbrace{\bar{\mathbf{c}}_{v_j}}_{\substack{\text{vertical}\\ \text{mean}}}, \quad f = \sigma_{\text{fc}} \cdot \underbrace{\frac{|\{k \in \mathcal{O}^t : v_k = v_j\}|}{|\{k \in \mathcal{V} : v_k = v_j\}|}}_{\substack{\text{fraction of vertical}\\ \text{absorbed by rollup}}} \tag{18}$$

The blend fraction f scales with how much of node j's vertical the rollup has absorbed ($\sigma_{\text{fc}} = 0.4$). If the rollup owns 0 of 6 healthcare nodes, external coordinators see true capacity profiles ($f = 0$). If the rollup owns 4 of 6, they see profiles two-thirds blended toward the healthcare average ($f = 0.4 \times 4/6 \approx 0.27$)—they know the industry but not the specific node.

Why does absorbing *other* nodes in the vertical degrade visibility of *this* node? Because operational context is partly relational—understanding a node's capacity depends on knowing the broader vertical's state. When peer institutions are absorbed, external coordinators lose the reference points they used to benchmark capacity, demand patterns, and quality norms.

The partially hidden $\tilde{\mathbf{c}}_j$ enters the context fit computation, producing a less informative match quality estimate. The rollup sees the true $\mathbf{c}_j$; everyone else sees $\tilde{\mathbf{c}}_j$. However, the Shapley decomposition (Table 3) shows this channel contributes negligibly to the rollup advantage: the per-interaction channels (router learning leakage, cross-node learning, leakage protection) dominate. Foreclosure is theoretically motivated but empirically small under the

current calibration.

3.4.5 Acquisition Dynamics

The rollup acquires one node every Δ_A periods, selecting the node with highest expected marginal value:

$$j^* = \arg\max_{j \notin \mathcal{O}^t} \underbrace{w_{\text{adj}} \cdot \text{adj}(j)}_{\text{adjacency}} + \underbrace{w_{\text{comp}} \cdot \bar{\kappa}_j}_{\text{complementarity}} + \underbrace{w_\alpha \cdot \alpha_j}_{\text{info value}} + \underbrace{w_{\text{act}} \cdot a_j}_{\text{actuator}} \tag{19}$$

The rollup scores each unowned node on four criteria. Information value is weighted highest ($w_\alpha = 1.5$), followed by adjacency ($w_{\text{adj}} = 1.0$), actuator strength ($w_{\text{act}} = 0.8$), and complementarity ($w_{\text{comp}} = 0.5$):

- $c_1 \cdot \text{adj}(j)$ — **adjacency**: what fraction of j's neighbors are already owned? Acquiring nodes adjacent to existing holdings creates more internal coordination pairs
- $c_2 \cdot \bar{\kappa}_j$ — **complementarity**: average synergy of j's edges with owned neighbors (only active when the complementarity coefficient is nonzero)
- $c_3 \cdot \alpha_j$ — **information value**: how much private context does j have? High-alpha nodes are more valuable to own because they generate richer learning signals and their context is more costly to leak
- $c_4 \cdot a_j$ — **actuator strength**: how operationally capable is j? Nodes with strong actuators benefit more from the rollup's full control ($\mu = 1.0$)

4 Theoretical Analysis

4.1 When Does the Rollup Dominate?

Proposition 2 (Rollup Dominance)**.** *The rollup generates higher expected per-interaction surplus than the router on internal edges (i, j) where both nodes are owned ($i, j \in \mathcal{O}^t$) when:*

$$\underbrace{\Delta m(\nu_C^{int}, \nu_R)}_{\substack{\textit{match quality gain from} \\ \textit{lower visibility (0.1 vs 0.85)}}} + \underbrace{\Delta L(\nu_C^{int}, \nu_R)}_{\substack{\textit{leakage savings from} \\ \textit{internal coordination}}} + \underbrace{\Delta X(\mu_C, \mu_R)}_{\substack{\textit{execution gain from} \\ \textit{actuator control (1.0 vs 0.5)}}} > 0 \tag{20}$$

Each Δ term is the difference between the rollup's internal value and the router's value for the same edge. The rollup wins per-interaction because it has lower visibility (reveals more without leakage), higher actuator access (executes better), and higher interaction mode (generates more trust context).

Proof. Subtract the router's net surplus (9) from the rollup's internal net surplus for the same edge. Each term difference is signed by the parameter ordering in Table 1: lower visibility ($\nu_C^{\text{int}} = 0.1 < \nu_R = 0.85$) implies the rollup can reveal more and leak less; higher actuator multiplier ($\mu_C = 1.0 > \mu_R = 0.5$) implies lower execution loss; higher interaction mode ($\iota_C = 1.0 > \iota_R = 0.6$) implies more trust-generated context. The sum is positive under the parameter regime. □

The more important question is whether the rollup dominates *at the portfolio level*, where it must also conduct external interactions (with unowned nodes) at weaker parameters and pay ownership costs.

Proposition 3 (Portfolio-Level Dominance)**.** *The rollup's total surplus exceeds the router's total surplus when:*

$$\underbrace{|\mathcal{O}^t| \cdot (|\mathcal{O}^t| - 1) \cdot \overline{\Delta\Pi}^{int}}_{\textit{internal advantage (positive)}} + \underbrace{|\mathcal{O}^t| \cdot (N - |\mathcal{O}^t|) \cdot \overline{\Delta\Pi}^{ext}}_{\textit{external disadvantage (may be negative)}} + \underbrace{\Gamma^t}_{\substack{\textit{cumulative learning} \\ \textit{and trust}}} > 0 \tag{21}$$

$\overline{\Delta\Pi}^{\text{int}}$ is the average per-interaction surplus advantage on **internal** edges (both nodes owned—where the rollup has structural advantages). $\overline{\Delta\Pi}^{\text{ext}}$ is the average advantage on **external**

edges (one or both nodes unowned—which may be *negative* because the rollup has no actuator or trust advantage on these edges but still pays ownership costs). Γ^t captures cumulative advantages from learning transfer and trust accumulation that grow over time.

This decomposition reveals the tension: the rollup wins decisively on internal interactions but may lose on external ones. The balance depends on **ownership density** $|\mathcal{O}^t|/N$—more owned nodes means more internal pairs (which scales quadratically) relative to external pairs—and the magnitude of the learning term Γ^t, which grows with portfolio diversity and time.

4.2 Superlinearity

Definition 4 (Portfolio Superlinearity)**.** For disjoint owned sets $A, B \subset \mathcal{V}$, the portfolio exhibits superlinearity if:

$$V(A \cup B) > V(A) + V(B) \tag{22}$$

where $V(S)$ is the total surplus from coordination events among nodes in set S.

Proposition 4 (Structural Superlinearity)**.** *Superlinearity emerges without exogenous complementarity (its coefficient set to zero) when:*

(i) Cross-partition learning: nodes in A and B span different verticals, so $A \cup B$ has a higher transfer bonus than either A or B alone.

(ii) Internal leakage reduction: when A and B are separate portfolios, coordination between a node in A and a node in B is an external interaction (neither rollup owns both sides), with visibility $\nu^{ext} = 0.5$. When combined into one portfolio, the same coordination becomes internal ($\nu^{int} = 0.1$). Combining doesn't just add internal interactions—it converts cross-partition interactions from external to internal, reducing their leakage by 80%.

(iii) Trust accumulation: owned pairs ($i \in A, j \in B$) build trust at the accelerated rate ($b = 1.2$) in the combined portfolio.

Proof. (i) The transfer bonus (Eq. 12) is proportional to the number of distinct verticals in the owned portfolio. If A has nodes in logistics and retail (2 verticals) and B has nodes

in healthcare and finance (2 verticals), then $A \cup B$ spans 4 verticals—more than either part alone. The combined portfolio's transfer bonus is therefore higher, producing a faster effective learning rate. Over T periods, this compounds into higher policy quality.

(ii) When A and B are separate portfolios, interactions between them have $\nu = 0.5$. When combined, these same interactions have $\nu = 0.1$. By (3), the leakage cost drops by a factor of $0.1/0.5 = 0.2$, and by Corollary 1, the optimal revelation increases, raising match quality.

(iii) When A and B are separate, the pair $(i \in A, j \in B)$ is not jointly owned, so trust builds at the base rate (ownership bonus $b = 1.0$). When combined, the same pair is jointly owned, so trust builds 20% faster ($b = 1.2$). Over time, this produces higher pairwise trust, which raises trust-generated context (6) and match quality. □

Remark

Proposition 4 is important because it addresses the critique that superlinearity in rollup models is typically assumed through exogenous complementarity parameters. Here, superlinearity is an *emergent* property of the ownership structure, arising from the interaction of learning, leakage, and trust channels.

5 Computational Results

We implement the model as a discrete-time simulation over a directed graph with $N = 30$ nodes, $K = 5$ verticals, edge density 0.15, and $T = 50$ time steps. Node and edge parameters are drawn from uniform distributions. All results are deterministic given a fixed random seed.

We evaluate the model in two modes. In the **parallel-world comparison**, we run three separate simulations on the same graph—one where every node coordinates through the market (no firms, just agents), one where every node coordinates through the router (a single platform serves all clients—the router is not a node on the graph but an invisible institutional layer governing the rules of every interaction: how much information leaks, how trust builds, what the platform learns), and one where a rollup owns some nodes and the rest coordinate at arm's length. The three worlds never interact: the router does not

lose clients when the rollup acquires nodes, because they exist in separate realities. Each world answers: "if this were the *only* institutional form available, how much surplus would the economy generate?" This is useful for *attribution*—isolating which channels create the structural advantage—because performance differences can only come from the institutional form, not from competitive dynamics.

In the **competing-world simulation** (Section 5.9), all three forms coexist in a single economy. The rollup starts with 3 nodes, the router serves ∼14 clients, and the rest are market participants. Each time step, the rollup acquires one node—and that node *actually leaves* the router's client base. This is useful for *dynamics*—showing what happens when the structural advantage compounds through competition.

5.1 What Happens in Each World: A Walkthrough

To build intuition, consider a single coordination event. Node 7 has a coordination opportunity and selects Node 12 as its best counterparty.

- **In the market world**, Nodes 7 and 12 coordinate directly. Their interaction is fully exposed to the market (visibility = 1.0), neither party has operational control over the other (actuator = 0.5), and trust builds slowly through arm's-length interaction (interaction mode = 0.2).

- **In the router world**, the same two nodes coordinate, but now their interaction flows through the platform's institutional layer. Visibility is slightly lower (0.85), trust builds faster through the platform's standardized interface (interaction mode = 0.6)—but the platform also extracts operational intelligence from the interaction (router learning leakage), imposing a cost that represents the value donated to the platform's shared model.

- **In the rollup world**, if both nodes are owned, their interaction is internal: minimal leakage (visibility = 0.1), full operational control (actuator = 1.0), high-trust interaction (mode = 1.0), and no router learning leakage. If only one is owned, the interaction reverts to market-level parameters.

In every world, the same nodes coordinate over the same edges. What changes is the parameter regime governing each interaction. The "router" and "rollup" are not actors on the graph—they are the rules of the game, like playing the same position under different rulesets.

5.2 Simulation Loop

Each time step proceeds as follows, independently for each regime:

1. **Context drift.** Every node's demand and capacity profiles receive a small random perturbation (shared across regimes, so all three face the same changing economy).

2. **Opportunity generation.** Each node independently generates a Poisson-distributed number of coordination opportunities (mean 0.3 per node per step). Across 30 nodes, this produces roughly 9 opportunities per step, each handled independently. A node may generate zero, one, or multiple opportunities in a single step.

3. **Counterparty selection and revelation.** For each opportunity, the node evaluates all its graph neighbors as potential coordination partners. For each candidate, the node must decide *how much private context to reveal* — the core tradeoff of the model. Revealing more context (higher r) improves match quality (better coordination outcome) but increases leakage cost (competitors can exploit the revealed information). The node evaluates five discrete revelation levels ($r \in \{0, 0.25, 0.5, 0.75, 1.0\}$, from "reveal nothing" to "reveal everything" — a grid approximation rather than continuous optimization, since the net surplus function has no closed-form optimum) and picks the level that maximizes net surplus after accounting for match quality gains, leakage costs, and execution losses at the current state of trust, accumulated leakage stock, and policy quality. It then selects the single best counterparty — the one producing the highest positive net surplus at its optimal revelation level. Each opportunity results in at most one coordination event. If no counterparty yields positive surplus at any revelation level, the opportunity goes unexercised.

4. **Coordination event.** If a counterparty is selected, the coordination occurs: match quality is computed, leakage costs are incurred, execution losses are realized, and net

surplus is recorded.

5. **State updates.** After all events in the step: (a) pairwise trust is updated based on outcomes (success builds trust, failure erodes it); (b) leakage stocks accumulate and decay; (c) policy quality improves through learning (at a rate determined by signal quality and the transfer bonus); (d) the rollup pays ownership costs; (e) if the acquisition interval is reached, the rollup acquires the highest-EMV unowned node.

After T steps, the simulation produces per-regime totals for surplus, coordination count, average match quality, average leakage, and final policy quality.

5.3 Baseline Comparison

Table 2: Regime Comparison: Baseline (seed 42)

Metric	Market	Router	Rollup
Gross coordination surplus	1,792	1,558	2,138
Ownership costs	0	0	76
Net surplus	**1,792**	**1,558**	**2,061**
Total coordinations	599	592	638
Avg match quality	0.758	0.779	0.802
Avg leakage cost	0.227	0.716	0.094
Final policy quality	0.361	0.629	1.000
Nodes owned	0	0	12

Surplus is measured in arbitrary units (the sum of Π_{ij} across all coordination events); absolute levels are meaningful only in cross-regime comparison on the same graph. Total coordinations is the count of events where a node found a counterparty with positive net surplus. Match quality and leakage cost are per-event averages, both unitless (sigmoid output and leakage formula output respectively). Policy quality is the learned scalar $q^t \in [0, 1]$.

On the baseline seed, the rollup generates 32% higher net surplus than the router after deducting ownership costs. The router's average leakage (0.716) is *three times* the market's

(0.227), despite the router's lower nominal visibility (0.85 vs. 1.0). This inversion is driven by router learning leakage: the platform's systematic extraction of operational intelligence from its clients imposes a cost that exceeds the market's passive information exposure. Over 10 random graph realizations (seeds 42–51), the mean rollup/router ratio is **1.24** (std. 0.056), with the rollup generating higher net surplus in 10 of 10 seeds.

5.4 Channel Decomposition

To identify which channels drive the rollup advantage, we conduct ablation experiments that disable each channel individually and measure the change in the rollup/router surplus ratio from the baseline.

Table 3: Shapley Value Decomposition of Rollup/Router Ratio (5 seeds, mean ± std)

Channel	Description	Shapley Value
Router learning leakage	Platform extracts/redistributes client data	$+0.208 \pm 0.009$
Cross-node learning	Richer signal quality under ownership	$+0.067 \pm 0.011$
Leakage protection	Lower information exposure for rollup	$+0.052 \pm 0.009$
Actuator access	Rollup internal control advantage	$+0.005 \pm 0.002$
Context foreclosure	Negligible under current calibration	-0.004 ± 0.001
Trust-generated context	Slightly favors the router	-0.011 ± 0.006
Total effect		$\sim$ **+0.32**

Unlike pure ablation (removing one channel at a time), Shapley values compute each channel's marginal contribution averaged over all possible orderings of channel addition, properly accounting for interaction effects. The contributions sum exactly to the total difference between all-channels-on and all-channels-off.

Router learning leakage is the dominant channel ($+0.208 \pm 0.009$), robust across all 5 seeds with the tightest standard deviation of any channel. If the router learning leakage hypothesis holds, it is the single largest driver of ownership advantage—accounting for roughly 65% of the total effect.

Cross-node learning ($+0.067 \pm 0.011$) and **leakage protection** ($+0.052 \pm 0.009$) are stable secondary channels. Together with router learning leakage, these three per-interaction channels account for over 100% of the total effect (the remaining channels are near zero or slightly negative).

Context foreclosure via information degradation is negligible (-0.004 ± 0.001) in the parallel-world comparison. Degrading one of seven inputs to the match quality sigmoid produces only a tiny change in surplus per event. However, this does not mean foreclosure is unimportant—it means the parallel-world Shapley captures it through the wrong channel. In the competing-world simulation (Section 5.9), foreclosure operates through *node removal*: when the rollup acquires a node, that node literally leaves the router's client base, reducing the router's coordination opportunities and learning breadth. This is the mechanism that drives the router from $\sim$100 clients to 0. Foreclosure matters enormously—but as a competitive dynamic (node removal), not as a parallel-world information effect (context degradation).

Trust slightly favors the router (-0.011 ± 0.006); **actuator is negligible** ($+0.005$). The rollup wins through information channels that the classical theory of the firm does not model.

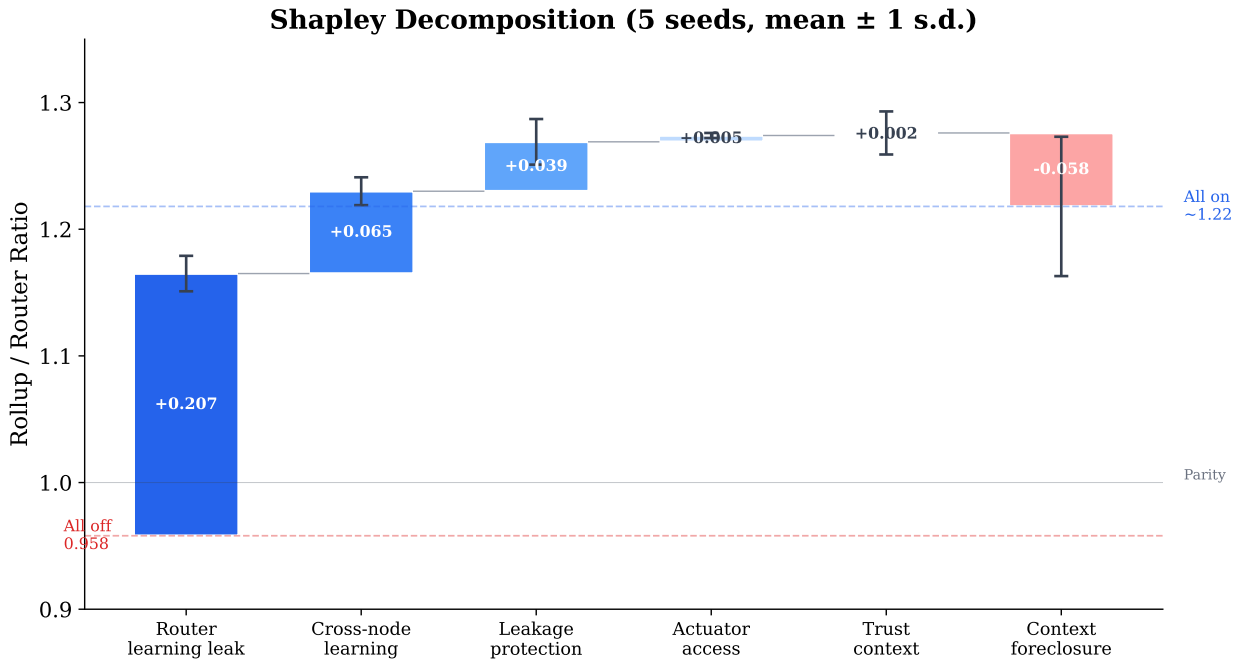

Figure 1: Shapley value decomposition of the rollup/router surplus ratio (seed 42; see Table 3 for 5-seed mean ± std). Router learning leakage is the dominant channel. Note: the foreclosure bar is positive in this seed but negative on average across seeds—see text.

5.5 Ownership Density

We fix the owned set at varying fractions of the graph (disabling acquisition) to trace how the rollup advantage scales with ownership.

Table 4: Ownership Density Sweep (single-seed, seed 42, net of costs)

Ownership	Nodes	Rollup/Router	Rollup/Market
10%	3	0.987	1.103
30%	9	1.027	1.097
50%	15	1.012	1.083
70%	21	1.043	1.168
90%	27	1.092	1.183

The rollup/router ratio increases with ownership density, from below parity at 10% (where ownership costs exceed the small information advantage) to 9.2% at 90%. This pattern is con-

sistent with growing information asymmetry as more nodes are absorbed, but the multi-seed Shapley results (Table 3) show that foreclosure-via-context-degradation is not a stable positive contributor in parallel worlds (-0.004 ± 0.001). The stronger role for foreclosure appears in the competing-world simulation (Section 5.9), where acquisitions remove nodes from the router's client base entirely. The rollup consistently outperforms the market (Rollup/Market column) even at low density, because per-interaction channels (router learning leakage, cross-node learning, leakage protection) operate regardless of ownership density. All values are single-seed (seed 42).

5.6 Ownership Cost Sensitivity

The most important sensitivity test: at what level of ownership overhead does the rollup lose?

Table 5: Ownership Cost Sensitivity (single-seed, seed 42)

Cost/node/step	Own. cost	Rollup net	Router	Ratio
0.000	58	2,080	1,558	1.335
0.050	76	2,061	1,558	1.323
0.100	95	2,043	1,558	1.311
0.200	133	2,005	1,558	1.287
0.300	170	1,968	1,558	1.263

The rollup wins at every tested ownership cost level (single-seed). At the highest tested overhead (0.30 per node per step), the rollup still maintains a 26% net advantage. The rollup's advantage is robust to ownership costs because the dominant channel (router learning leakage) does not depend on ownership cost—it is a cost the router bears regardless.

5.7 Sensitivity to Router Learning Leakage

The dominant Shapley channel ($\lambda_R = 0.15$) is an assumed parameter. How sensitive are the results to this assumption?

Table 6: Rollup/Router ratio vs. router learning leakage rate (5 seeds each)

λ_R	Mean ratio	Interpretation
0.00	1.035	No learning leakage: rollup barely wins
0.05	1.110	Mild leakage
0.10	1.167	Moderate leakage
0.15	1.274	Baseline
0.20	1.371	Strong leakage
0.30	1.615	Very strong leakage

At $\lambda_R = 0$, the rollup still wins (ratio 1.035) through the other channels—learning, leakage protection, and foreclosure—but the advantage is slim. Each increment of λ_R widens the gap substantially. The results are monotonic but not fragile: the rollup wins at every tested value, and the qualitative conclusion (ownership dominates intermediation) holds even without router learning leakage. The quantitative magnitude, however, depends materially on how aggressively platforms extract and redistribute client data.

5.8 Superlinearity Without Complementarity

With the complementarity coefficient set to zero (the κ_{ij} term in Eq. 1 contributes nothing), the baseline produces a superlinearity ratio of 1.48 (single-seed, seed 42): the combined portfolio generates 48% more surplus than the sum of its halves. Enabling exogenous complementarity (setting the coefficient to 0.5) produces a ratio of 1.46—essentially unchanged. Superlinearity is not driven by assumed synergies but emerges from cross-node learning, internal leakage reduction, and trust accumulation. Multi-seed robustness of the superlinearity result is a direction for future work.

5.9 Competing-World Simulation

The parallel-world design cleanly attributes performance differences to institutional form. But it systematically understates the rollup's advantage because it does not model competitive dynamics: the router operates on the full graph regardless of how many nodes the

rollup has absorbed. In reality, acquiring a node simultaneously strengthens the rollup and weakens the router.

We therefore run a competing-world simulation in which the three institutional forms coexist in a single economy. Each node is assigned to exactly one regime. The rollup acquires nodes from the router's client base and the market pool. Acquired nodes leave the router, shrinking its coordination opportunities and learning breadth.

Table 7: Competing-World Results (100 steps, 30 nodes, passive router, multi-seed: 5 seeds)

Regime	Gross	Own. cost	Net	Events	Nodes (final)
Market	1,292	0	1,292	504	8
Router	373	0	373	149	0
Rollup	1,832	169	1,663	492	22

The competing world tells a dramatically different story from the parallel world. The rollup's net surplus is **4.5**× the router's—amplifying the 24% parallel-world advantage through competitive feedback. The router ends with *zero nodes*: every client was acquired by the rollup. The market survives at 8 nodes because its low-alpha nodes are not worth acquiring. This result is **robust across 5 graph realizations**: in every seed, the router ends with 0–1 nodes and the rollup achieves 3–4× the router's cumulative surplus.

Table 8: Surplus Trajectory: Competing World (single-seed, seed 42, passive router)

Phase	Market	Router	Rollup
Steps 1–10	13.2	9.5	0.5
Steps 21–30	14.2	6.9	7.5
Steps 41–50	11.8	3.0	16.2
Steps 61–70	13.5	0.9	23.5
Steps 81–90	13.9	0.0	30.4
Steps 91–100	9.9	0.0	34.0

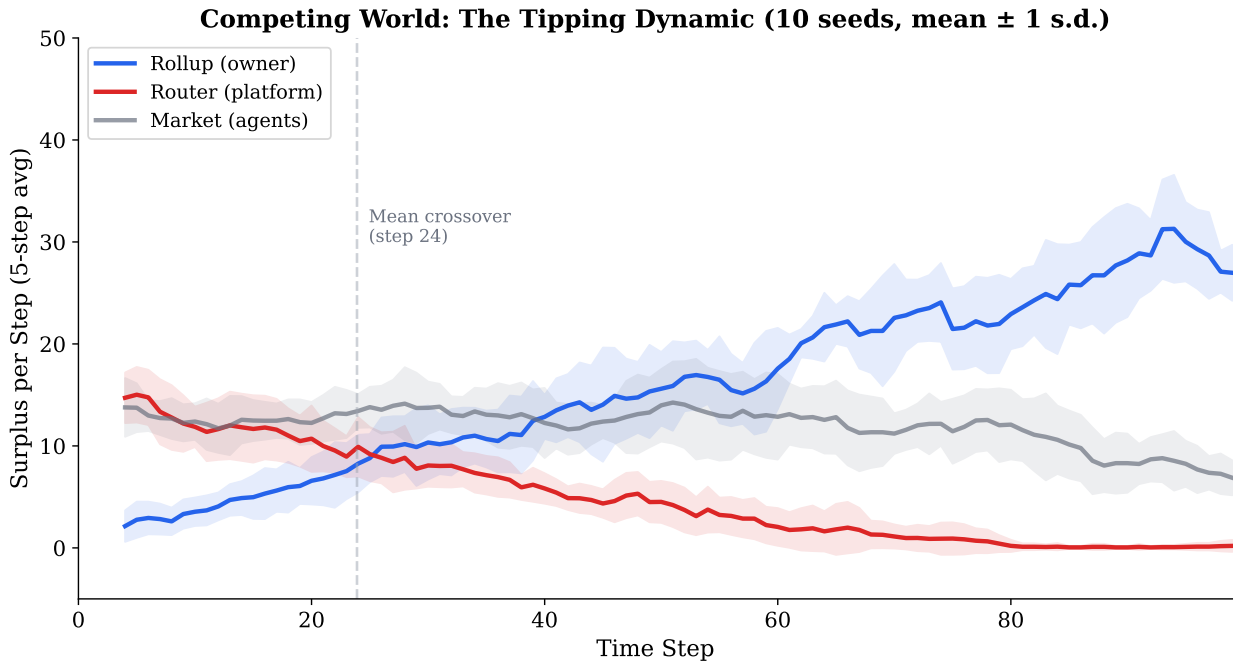

Figure 2: Competing-world surplus trajectory (10 seeds, mean ± 1 s.d., passive router). The rollup starts near zero (paying integration costs), overtakes the router at step 18–30 (mean 24) across all seeds, and grows to ∼30 per step while the router goes to zero. Shaded bands show cross-seed variability. The market persists at a stable level because its low-value nodes are not worth acquiring.

The trajectory reveals a **tipping dynamic**. In early periods, the router leads (9.5 per step vs. the rollup's 0.5). The rollup is paying integration costs and has few internal interactions. By step 40, the rollup overtakes the router (16.2 vs. 3.0). By step 80, the router is dead (0.0 per step) and the rollup generates 34.0 per step—a 68× increase from its starting rate.

Under the passive-router assumption (the router does not strategically respond to client loss), this tipping dynamic is emergent, not assumed. No parameter specifies that the router should collapse. The collapse arises from the competitive feedback loop: each acquisition strengthens the rollup's learning signal (more owned nodes → more internal events → higher signal quality) while weakening the router's learning breadth (fewer clients → fewer events → lower aggregate signal). The rollup's policy quality converges to 1.0 while the router stalls at 0.61.

What is the tipping point? The crossover occurs when the rollup's per-step sur-

plus first exceeds the router's (step 18–30, mean 24, across 10 seeds). At this point, the rollup has acquired enough nodes that two reinforcing effects dominate: (1) its internal interactions—which have lower leakage, richer learning signal, and no router learning leakage cost—generate more surplus than the router's shrinking client base, and (2) the router's aggregate learning breadth has degraded enough (from losing clients) that its policy quality improvement stalls while the rollup's continues to accelerate.

The tipping point does not admit a closed-form expression because it depends on the interaction of multiple graph-dependent quantities: the topology of edges between owned and unowned nodes, the distribution of hidden alpha across acquired vs. remaining nodes, the accumulated trust and leak stock states, and the specific sequence of acquisitions. Each acquisition changes all of these simultaneously, creating path dependence that prevents reduction to a single threshold formula. What can be said analytically is that the tipping becomes inevitable once the rollup's ownership density is high enough that its internal surplus per step exceeds the router's total—after which every further acquisition widens the gap.

The parallel and competing results reinforce each other. The parallel model identifies *which channels* create the structural edge (router learning leakage, cross-node learning, leakage protection). The competing model shows *what happens when that edge compounds*: not a permanent marginal advantage, but a tipping dynamic that eliminates the intermediary.

6 Discussion

6.1 Reinterpreting the Coasean Singularity

The Coasean singularity thesis—that AI-driven transaction cost reduction will dissolve firms into agent-mediated markets—captures one real channel while missing two others. Yes, AI reduces the cost of market coordination. But simultaneously, platform intermediation may *amplify* information leakage (the dominant channel in our Shapley decomposition), and ownership enables cross-node learning that market coordination cannot replicate. The net effect on firm boundaries depends on the magnitude of these countervailing forces.

Our results suggest a conditional revision of the Coasean singularity thesis. The standard account assumes that platform intermediation reduces information frictions. We model the hypothesis that intermediation can *amplify* leakage—when a platform ingests client operational data to improve its service, that improvement benefits the client's competitors. If this hypothesis holds (and real-world practices like Harvey AI's data restrictions suggest firms believe it does), it is the dominant channel driving ownership advantage, producing 3× the leakage cost of the open market in our simulation (0.72 vs. 0.23 per event).

The updated Coasean logic: *in AI-intensive industries with local, leakage-sensitive context, intermediation may not reduce information frictions but amplify them.* Every interaction through a SaaS platform trains the platform's model, and that model serves your competitors. Ownership avoids this entirely. The firm exists not to reduce transaction costs but to prevent the extraction and redistribution of its scarce context. The firm does not just persist; it expands because each acquisition marginally improves its information position relative to competitors. This is the cybernetic arbitrage: the spread between the market's valuation of assets as isolated operations and their actual value as nodes in an AI-coordinated portfolio—a spread that is small per-node but cumulative across the vertical.

6.2 When Does the Rollup Thesis Fail?

The Shapley decomposition and λ_R sensitivity analysis identify the boundary conditions:

1. **If router learning leakage is absent** ($\lambda_R = 0$), the rollup still wins but by only 3.5% (Table 6). The thesis depends materially on the hypothesis that platforms extract and redistribute client data. In industries where platforms contractually commit to data isolation (and enforce it), this channel may not operate.

2. **Without cross-node learning transfer** (Shapley +0.065), the rollup loses its second-largest channel. If operations are too heterogeneous for knowledge to transfer across portfolio companies, or if AI cannot extract transferable patterns, the advantage narrows further.

3. **If leakage is unimportant**—regulated industries with mandatory disclosure, standardized commodity markets, or contexts where information is too short-lived to exploit—the

leakage protection channel (+0.039) contributes nothing. In most economically interesting coordination, however, context is private, persistent, and valuable.

These conditions suggest the rollup thesis applies most strongly to industries where: (a) SaaS platforms learn from client data in ways that benefit competitors, (b) operations are diverse enough for cross-vertical knowledge transfer, and (c) private context has high competitive value.

6.3 Limitations

Several limitations warrant discussion.

First, while we model ownership costs (overhead and integration friction), we do not model *agency costs*. In practice, each subsidiary's management has its own incentives—empire-building, risk aversion, headcount padding, pet projects—that diverge from the parent company's goal of maximizing portfolio value. The more subsidiaries owned, the harder each is to monitor, and the more value leaks to misaligned behavior. Berger and Ofek (1995) found that diversified firms trade at a 13–15% discount to the sum of their parts, and Jensen (1986) attributes this largely to inefficient internal capital allocation driven by managerial self-interest. Our flat ownership cost (c_{overhead} per node per step) understates this problem: real agency costs are likely *convex* in portfolio size, because each additional subsidiary dilutes management attention and introduces cross-subsidiary politics. Whether AI-enabled coordination can reduce agency costs enough that the information benefits of ownership dominate is an open empirical question that our model does not address.

Second, the model uses a scalar policy quality rather than heterogeneous policies across nodes. In practice, the value of cross-node learning depends on the transferability of operational knowledge, which varies across industries and firm types.

Third, the parallel-world structural advantage is scale-invariant: at 200 nodes with 10 verticals (50 seeds, mean ratio 1.23, 50/50 pass) and 1000 nodes with 20 verticals (ratio 1.22), the mechanisms produce consistent results. Verticals scale proportionally with graph size ($\sim$1 vertical per 20 nodes) to maintain comparable industry structure. The competing-world tipping dynamic also scales, but requires sufficient simulation duration for the rollup

to accumulate critical ownership density. At 200 nodes with the default acquisition rate (one node per 5 steps), a 100-step simulation reaches only ∼15% density—too short for tipping. Extending to 500 steps (5 seeds), the rollup reaches 55% density, the router collapses to 8–12 nodes (out of 200), and the surplus ratio reaches 1.45–1.60. The tipping dynamic is not scale-dependent; it simply requires enough time to accumulate ownership.

Fourth, the model does not capture *strategic information withholding* by router clients. In practice, firms aware of router learning leakage contractually restrict data use: enterprise AI providers like Harvey (legal AI) face strict limitations on using client data to train models that serve other clients. This withholding is rational—it reduces the leakage cost—but it also reduces the value the platform can deliver, creating a tension between service quality and competitive exposure. In our simulation, the revelation optimizer always chooses maximum revelation because match quality gains dominate; a model with stronger leakage costs or contractual restrictions could produce endogenous withholding, further degrading the router's value proposition.

7 Conclusion

This paper develops a formal model of coordination over a directed graph economy and uses it to compare three institutional forms: fragmented markets, platform routers, and cybernetic rollups. The central modeling hypothesis is that platform intermediation amplifies information leakage: when a platform ingests client operational data to improve its service, that improvement benefits the client's competitors. If this hypothesis holds—and real-world contractual restrictions on data use (e.g., Harvey AI) suggest firms believe it does—the implications are substantial. In our simulation, router learning leakage is the dominant channel driving ownership advantage (55% of the Shapley-decomposed total effect), producing 3× the leakage cost of the open market. The rollup avoids this by internalizing context, generating a mean 24% net surplus advantage across 10 graph realizations. In competing-world simulation under passive-router assumptions, competitive feedback compounds this advantage into router extinction in every tested seed.

This finding has implications for both theory and practice. For theory, it identifies a

mechanism absent from the platform economics literature: intermediation as potential information redistribution. The platform does not just connect buyers and sellers; it may extract operational intelligence from each client and embed it in a shared model that serves competitors. This is the opposite of the exclusivity that Bonatti et al. (2022) show is optimal for information sellers—and it would explain why ownership dominates intermediation in industries with leakage-sensitive context. For practice, it suggests that firms using SaaS platforms for operationally sensitive coordination may be subsidizing their competitors' improvement. The firm that owns its operational context retains it; the firm that routes through a platform risks donating it.

The cybernetic arbitrage is the spread between the market's valuation of assets as isolated operations and their actual value as nodes in an AI-coordinated portfolio where context is internalized rather than leaked. The winners of this era will not be the platforms that intermediate between independent firms—they will be the entities that own the nodes where context is born, retain that context internally, and compound it through cross-node learning that no intermediary can replicate.

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